



Institutional deterioration and entrepreneurial investment: The role of political connections



Jianhua Ge^a, Laura J. Stanley^b, Kimberly Eddleston^{c,*}, Franz W. Kellermanns^{b,d}

^a School of Business, Renmin University of China, China

^b Belk College of Business, University of North Carolina at Charlotte, United States

^c D'Amore-McKim School of Business, Northeastern University, United States

^d WHU - Otto Beisheim School of Management, Germany

ARTICLE INFO

Article history:

Received 10 June 2016

Received in revised form 26 March 2017

Accepted 24 April 2017

Available online 3 May 2017

ABSTRACT

Although many start-ups struggle to grow partly due to institutional voids in transitioning economies, empirical observations have contradicted this dominant view in the literature in that even with deficient formal institutions many economies (e.g., China) have high rates of entrepreneurship in recent years. To understand this paradox, we propose a politically contingent view of the relationship between the institutional environment and entrepreneurs' reinvestment in their business. Our empirical study of entrepreneurs in China shows that the impact of institutional deterioration on entrepreneurial reinvestment substantially hinges on entrepreneurs' political connections. As such, institutional deterioration does not reduce entrepreneurial reinvestment for all entrepreneurs; rather, when entrepreneurs have political connections they are willing to reinvest in their business despite a weakening institutional environment. Our framework suggests that in an environment perceived as harsh to business, political connections can encourage entrepreneurs to see opportunities for growth. In contrast, entrepreneurs that lack political connections will mainly see threats in a deteriorating institutional environment and thus, limit their business reinvestment.

© 2017 Elsevier Inc. All rights reserved.

Executive summary

Research on institutions and entrepreneurship has long argued that “a functional business environment provides positive incentives for entrepreneurs while a weak one is likely to be deleterious” (Estrin et al., 2013 p. 566). However, recent empirical studies, especially those on transitioning economies, have found that even with deficient formal institutions many economies have high levels of entrepreneurial activities and entrepreneurship investment (McMillan and Woodruff, 2002; McMillan and Woodruff, 1999; Zhou, 2013). It is therefore puzzling that in economies like China that are characterized by weak legal infrastructures, high levels of corruption and regulation, and insufficient market intermediaries, that entrepreneurial activities are not as low as expected.

In fact, entrepreneurship in China has been referred to as a paradox since despite having an institutional environment that is unfriendly to business, during the last two decades the country has had one of the world's highest entrepreneurship rates (Ebner, 2014; Eunni and Manolova, 2012; Huang, 2010). Economists are puzzled by how a transitioning economy that supports market

* Corresponding author.

E-mail addresses: gejianhua@rbs.ruc.edu.cn (J. Ge), lstanl11@unc.edu (L.J. Stanley), k.eddlestone@neu.edu (K. Eddleston), kellermanns@unc.edu (F.W. Kellermanns).

socialism can produce such a vibrant, entrepreneurial community. They are further puzzled by why entrepreneurship continues to grow in Chinese provinces with a deteriorating institutional environment.

Our study seeks to help solve this puzzle by going beyond the predominant view in the literature that treats institutions as exogenous to entrepreneurship, and instead acknowledging that entrepreneurial opportunities can be institutionally constructed. For example, not only do political and regulative institutions offer rules for the economic market that can spur entrepreneurship, but they themselves should be seen as products in the political marketplace (Buchanan, 1987). Thus, rather than assume that entrepreneurship activities are only motivated by opportunities stemming from economic markets, we argue that opportunities can also be reaped from political markets, especially within those economies where government significantly affects business (Bonardi et al., 2005; Oliver and Holzinger, 2008). Because firms compete in political markets, their connections with political authorities may play an important role in producing opportunities and motivating entrepreneurial reinvestment. Accordingly, we propose a politically contingent model of institutional deterioration and entrepreneurial reinvestment that proposes that institutional deterioration is not universally perceived as a threat that undermines entrepreneurial reinvestment. Rather, when entrepreneurs have political connections that can endow advantages under weak institutions, institutional deterioration can be perceived as an opportunity that promotes entrepreneurs' reinvestment. Our study therefore sheds light on China's entrepreneurship paradox by demonstrating how political connections alter the effect of the institutional environment on entrepreneurs' business reinvestment.

Using national survey data of Chinese entrepreneurial firms, our study reveals a negative relationship between institutional deterioration and reinvestment rate for nonpolitically-connected entrepreneurs. In a striking contrast, institutional deterioration positively affects reinvestment rate for politically-connected entrepreneurs. These results suggest that institutional deterioration is perceived substantially differently (i.e., as offering opportunities or threats) by politically-connected entrepreneurs and those without political connections. Accordingly, political connections appear to have a distinct impact on entrepreneurs' reinvestment in their business in transitioning economies like China.

This study offers several contributions. First, our politically contingent view of the relationship between the institutional environment and business reinvestment in transitioning economies identifies an important boundary condition to our current understanding of the impact of institutions on entrepreneurship. Second, applying research on political ties to entrepreneurship, this study highlights how entrepreneurs' decisions and activities are not solely determined by opportunities in their economic market, but also hinge on their political power and the resources gained from political markets. Due to their heterogeneity in political capital, entrepreneurs are stratified in the political market where their relative advantages or disadvantages will be ultimately channeled into their business. Third, prior empirical analyses on institutions and entrepreneurship mainly use country-level data that shadow the within-country variations and the heterogeneity among entrepreneurial firms. This study directly tackles these issues by capturing the entrepreneurs' perception of the institutional environment. Therefore, our study draws attention to how the political arena can create and sustain advantages for entrepreneurs, thereby explaining differences in how entrepreneurs react to a deteriorating institutional environment.

1. Introduction

Entrepreneurship in China has been characterized as a *paradox*. Although the rule of law, market competition, and business environment are viewed as unfriendly to business, over the last two decades the country has had one of the world's highest rates of entrepreneurship (Ebner, 2014; Eunni and Manolova, 2012; Huang, 2010). Economists are puzzled by how a transitioning economy that supports market socialism can produce such a vibrant, entrepreneurial community. This paradox has led to much interest in studying China's startups (Pistrui et al., 2001; Yang and Li, 2008) and the role of its institutional environment in affecting firm performance (Banalieva et al., 2015; Kshetri, 2007; Peng and Jiang, 2010). It has also led to calls for research to investigate how a weak institutional environment affects entrepreneurs' reinvestment in their business – defined as the percentage of after-tax profits that were reinvested – since the prosperity and growth of SMEs is seen as critical to alleviating poverty and creating jobs in transitioning economies (Bruton et al., 2010; George et al., 2015).

While research has long studied the relationship between institutions and entrepreneurship, this line of inquiry has been reinvigorated in recent years as scholars recognize how a vibrant entrepreneurship community is necessary in developing a middle class and meeting the needs of underserved individuals (Bruton et al., 2013; Hwang and Powell, 2005; Jennings et al., 2013; Puffer et al., 2010; Valdez and Richardson, 2013). Regarding the impact of institutions on entrepreneurship, a long-standing assumption is that “a functional business environment provides positive incentives for entrepreneurs while a weak one is likely to be deleterious” (Estrin et al., 2013, p. 566). Such research assumes that a weak institutional environment forces entrepreneurs to focus more on survival than growth (Smallbone and Welter, 2001). However, recent studies focusing on transitioning economies have found that even with deficient formal institutions, many economies have high levels of entrepreneurial investment and entrepreneurship activities (McMillan and Woodruff, 2002, 1999; Zhou, 2013). Additionally, research tends to assume that the institutional environment is improving in transitioning economies like China, although this is actually not the case (Banalieva et al., 2015). Some provinces in China are experiencing reform reversals and institutional deterioration that are likely to generate uncertainties of rewards from entrepreneurial investment and threaten entrepreneurs' growth aspirations (Banalieva et al., 2015; G. Fan et al., 2007; J.P.H. Fan et al., 2007). It is therefore puzzling that in Chinese provinces with a deteriorating institutional environment characterized by weak legal

infrastructures, high levels of corruption and regulation, and insufficient market intermediaries, that entrepreneurship activities are not as low as expected.

China's paradox leads us to reflect on the predominant view in the literature. While extant research tends to treat institutions as exogenous to entrepreneurship, entrepreneurship opportunities can be institutionally constructed. For example, political and regulative institutions not only offer rules for the economic market that can breed entrepreneurship, they themselves are products in the political marketplace (Buchanan, 1987). However, studies tend to focus on how entrepreneurship activities are motivated by opportunities stemming from economic markets (i.e., substitute goods/services), while ignoring those that can be reaped from political markets (i.e., rival political and policy issues). Such a gap in the literature is surprising since firms often compete in political markets for scarce resources and opportunities (Claessens et al., 2008; Faccio, 2006). Indeed, it has been argued that the political arena is underestimated as a means of creating and sustaining a competitive advantage (Lawton et al., 2013). This gap in the literature is particularly surprising in the context of many transitioning economies where the government significantly affects business and political authorities are not only setting the rules but are also active players in the economy through their control of resources and opportunities as well as state-owned enterprises (Bonardi et al., 2005; Oliver and Holzinger, 2008).

In addition, most studies assume that firms are homogeneously affected by institutions, yet, firms' linkages with institutional authorities (e.g., state, government, regulators) are diverse and heterogeneous (Powell, 1991; Scott, 2014). Political connections refer to relationships between an entrepreneur and political institutions or agents such as party leaders, government officials, and elected legislators (Faccio, 2006; Fisman, 2001; Siegel, 2007). Close connections with political authorities can potentially benefit firms in accessing scarce resources, private information, and favorable policies (Claessens et al., 2008; Faccio, 2006; Fisman, 2001). Political connections also help entrepreneurs obtain protection of property rights (Chen et al., 2011; Zhou, 2013, 2014) which are necessary in encouraging entrepreneurs to reinvest in their business (North, 1990). The existence of political connections may therefore help explain why some entrepreneurs reinvestment in their business despite a deteriorating institutional environment.

Accordingly, in this study we investigate how political connections alter how entrepreneurs respond to the institutional environment of a transitioning economy by proposing a politically contingent model of the relationship between institutions and entrepreneurial reinvestment. Specifically, we investigate how institutional deterioration, which refers to the increasing weakness and inefficiency of the institutional environment (Batjargal et al., 2013), affects entrepreneurs' reinvestment in their business. While the predominant institutional view suggests a negative effect between institutional deterioration and entrepreneurial reinvestment (Ebner, 2014; Eunni and Manolova, 2012; Huang, 2010), we argue that political connections play an important moderating role in this relationship. That is, political connections appear to be an important source of social capital in China given the prominence of government intervention (Banalieva et al., 2015; Peng and Jiang, 2010; Zhou, 2014) as well as the cultural norm of *guanxi*—interpersonal relationships that facilitate the exchange of favors, such as those between entrepreneurs and politicians—in facilitating business (Peng, 2004; Puffer et al., 2010; Xin and Pearce, 1996). As such, for those entrepreneurs in China with political connections, a deteriorating institutional environment may not impede their business reinvestment in comparison to those without political connections, since their political ties can help fill institutional voids and create a sense of security, stability and opportunity that lead them to increase their business reinvestment.

Our study makes at least three contributions to the literature. First, in studying entrepreneurial business reinvestment in a transitioning economy, we suggest that entrepreneurial opportunities can stem from political markets and thus consider the role of political connections in modifying entrepreneurs' reaction to a deteriorating institutional environment. We therefore propose a politically contingent model of the relationship between the institutional environment and business reinvestment that helps explain why some entrepreneurs invest in their business despite a deteriorating institutional environment while others significantly limit that investment. Second, utilizing a national survey of SME owners in China, we explore how their perception of the institutional environment affects their business reinvestment. This varies from the majority of other studies in China that have focused on publicly listed companies and macro-level measures of pro-market reforms and institutional indices (Banalieva et al., 2015; Peng and Jiang, 2010). As such, rather than assuming that all entrepreneurs react similarly to their institutional environment, our study captures the degree to which entrepreneurs perceive their institutional environment as becoming better or worse for conducting business. Finally, instead of studying the rate of entrepreneurship in a transitioning economy as most research has done (Yang and Li, 2008; Yu and Stough, 2005), we focus on entrepreneurs' willingness to reinvest in their business (Cull and Xu, 2005; Johnson et al., 2002; Zhou, 2016, 2013). Entrepreneurial investment including reinvestment is central to the process of entrepreneurship and largely affects entrepreneurial survival and growth. The decision to reinvest in one's business is among the most important decisions entrepreneurs can make since it directly affects the expansion or shrinking of the business (McCarthy et al., 1993). Such a focus is important to entrepreneurship research in transitioning economies since the growth of SMEs is critical to developing prosperity and moving beyond necessity-driven entrepreneurship to opportunity-driven businesses (Bruton et al., 2013; George et al., 2015).

Below, we provide an overview of entrepreneurship in transitioning economies focusing on China and propose that entrepreneurs' perception of the institutional environment influences their rate of business reinvestment. We then discuss the role of political connections to entrepreneurs and propose a politically contingent model of the relationship between the institutional environment and business reinvestment. This is followed by our study methodology and presentation of results and discussion.

2. Theory and hypotheses

2.1. Entrepreneurship in a transitioning economy

Entrepreneurship and private firm ownership is a relatively new phenomenon in transitioning economies like China (Puffer et al., 2010; Zhou, 2014). Transitioning economies are those which were once under central planning by the government and are now transitioning into market-based economies (Kriauciunas and Kale, 2006). Transitioning economies use various types and levels of pro-market reforms to decentralize and limit state control in the market, privatize property rights, reduce industry entry barriers and minimize government intervention in resource allocation (Park et al., 2006). At the same time, not all economic transitions are accompanied by political transformations toward democracy. In countries like Russia and China, the political regime remains authoritarian (Whitley, 1999). With regards to China in particular, the Communist Party retains tight control over both political and economic arenas. “While gradualism in liberalizing the market and privatizing state assets...the delay in granting full rights to private entrepreneurs purely reflects ideological rigidity and institutional inertia against changes” (Peng, 2004, p. 1054). As such, the rule systems are still weak and the political uncertainties surrounding business are relatively high (Haveman et al., 2017; Zhou, 2016). Moreover, the country is organized into relatively autonomous and institutionally different provinces/regions, which has produced a great degree of institutional variation within the country (Banalieva et al., 2015). Although pro-market reform policies are set by the Chinese federal government, they are implemented at various levels and speeds by the provincial governments.

Before pro-market reforms, which began in 1978, private enterprise was forbidden in China. Although the environment progressed during the 1990's and early 2000's, entrepreneurship was still discouraged due to a reluctance to reform legal and market institutions that would support entrepreneurship (Zhou, 2014). Yet, by 2005, China had approximately 24 million private enterprises (Loyalka, 2006) and in 2012, the number of registered SMEs exceeded 4.3 million (Ministry of Commerce People's Republic of China, 2012). These SMEs contribute 58.5% of GDP, 50% of tax revenues, and 75% of new jobs each year. SMEs also account for 99% of China's registered enterprises (Ministry of Commerce People's Republic of China, 2012). However, in comparison to other economies, the institutional environment in China remains ridden with barriers and obstacles for entrepreneurs. According to the Economic Reform Report that assesses the role of political, economic, and government factors in promoting entrepreneurship, China is one of the lowest ranked countries, ranking 138 out of 184 economies. In fact, it has been stated that “from a ‘quality of government’ perspective, the regulatory environments of the economies in transition from central planning and dominant communist rule such as...China appear to be the most unfavorable for the promotion of entrepreneurship” (Eunni and Manolova, 2012, p. 176).

It is widely acknowledged that entrepreneurship is influenced by institutions (North, 1990). Because they affect almost all entrepreneurial processes (i.e., information acquisition, economic foresight, risk tolerance, property rights, financing and market entry) (Fogel et al., 2006), the institutional environment is expected to influence entrepreneurs' growth aspirations and willingness to reinvest in their business (Estrin et al., 2013; Zhou, 2013). In turn, the dominant view in the literature is that a well-functioning institutional environment will facilitate entrepreneurship, while a weak one will hamper it (Banalieva et al., 2015; Bruton et al., 2010; Cuervo-Cazurra and Dau, 2009). Indeed, using the Global Entrepreneurship Monitor (GEM) data drawn from both developed and developing countries, McMullen et al. (2008) found that strong property rights, fiscal freedom and monetary freedom foster entrepreneurial activities. Similarly, a study by Fogel et al. (2006) that captured the institutional environment through factors including property rights, legal regime and government quality, suggested that more market-based institutions boost entrepreneurship. Studies have also shown that entrepreneurs will invest more of their profits if the institutions can secure the private returns from their investment activity (e.g., Cull and Xu, 2005). These findings support the institution-based view of the firm that emphasizes that institutions matter in promoting growth since they “support the effective functioning of the market mechanism such that firms and individuals can engage in market transactions without incurring undue costs or risks” (Meyer et al., 2009, p. 7).

The detrimental impact of hostile institutional environments on entrepreneurship has also been closely examined. Weak property rights generate uncertainty that makes long-term investment risky and destroys the “transactional trust” that is essential to entrepreneurship (Fogel et al., 2006). Ambiguous and inordinate amounts of legislation also increase transaction costs and decrease operational efficiency. Furthermore, cumbersome regulation and excessive government intrusion raise the threat of arbitrary expropriation and generate additional costs of new business operations (Estrin et al., 2013). In turn, these threats and potential losses incubated in weak institutions greatly hinder business growth (Puffer et al., 2010; Zhou, 2014). Risk and uncertainty due to high volatility in an institutional environment makes it harder for entrepreneurs to predict parameters necessary to make strategic decisions and to invest in growth (Wright et al., 2005; Xu and Meyer, 2013). For example, entrepreneurs in institutional environments where corruption is high and government is large have lower growth aspirations (Estrin et al., 2013). From a survey of entrepreneurial firms in several post-communist countries, Johnson et al. (2002) also found that weak property rights discourage firms from reinvesting their profits.

Institutional voids related to financial credit and property rights protection have also hurt the business growth of SMEs in China (Puffer et al., 2010; Zhou, 2014). The implementation of laws protecting property rights are inconsistent and unpredictable in China and thus, citizen's private property is not fully protected from expropriation by the state (Eunni and Manolova, 2012). Yet, the protection of private property rights is a precondition for entrepreneurs to reinvest in their business; that is, entrepreneurs will not reinvest in their business if they do not expect to keep the fruits of their labor (North, 1990). Indeed, due to a

lack of protection for private property, Chinese entrepreneurs appear reluctant to reinvest in their business (Zhou, 2009). Given these considerations, according to the dominant view in the literature, we propose the following baseline hypothesis:

Hypothesis 1. Institutional deterioration has a negative impact on entrepreneur's reinvestment in their business.

2.2. Entrepreneurs' political connections

While research typically assumes that good institutions are conducive to entrepreneurship and weak institutions are detrimental, the reality is mixed. For example, according to Estrin et al. (2013), protection of intellectual property rights does not always impact entrepreneurship. Stenholm et al. (2013) also suggested that the regulatory dimension of country-level institutional arrangements (e.g., economic freedom and property rights) had no substantial effect on entrepreneurial aspirations. Contrary to expectations, Valdez and Richardson (2013) found that regulative institutional support did not positively affect entrepreneurial activity, rather, their results revealed some negative effects on entrepreneurial activities. Findings like these suggest that the effect of the institutional environment on entrepreneurship is complex and that boundary conditions should be identified and explored. Indeed, scholars have shown that in weak institutions, social mechanisms (e.g., social networks, kinship networks) can be employed to buffer institutional deficiencies that depress entrepreneurship (Estrin et al., 2013; Peng, 2004; Puffer et al., 2010). In China where political authorities not only shape the institutional environment but also participate in the economy through the command of state-owned enterprises and the control of resources and opportunities, those without political connections have difficulties gaining reliable access to scarce resources and opportunities (Haveman et al., 2017; Park et al., 2006). Thus, we suspect that in transitioning economies where the government and political authorities still play a significant role in shaping the business environment, the relationship between institutions and entrepreneurial reinvestment is likely to be contingent on entrepreneurs' political connections.

Although the strategic management literature has recognized how corporate political activity such as lobbying and campaign contributions can benefit firm performance (Hillman et al., 2004; Hillman and Hitt, 1999; Lux et al., 2011), little entrepreneurship research has explored how political connections can benefit SMEs. Political connections can potentially benefit a business as firms also compete in *political* markets (Lawton et al., 2013). Parallel to the exchange between consumers and goods/services suppliers in the economic market, the political market concerns the exchange between entrepreneurs and political actors, including politicians, bureaucrats, and legislators (Bonardi et al., 2005). The political intermediate products generated by governments (e.g., permission to trade and invest, protection against sovereign risk) do not just exist; they are created in the context of an enacted environment that is shaped through political actions (Boddewyn and Brewer, 1994). Furthermore, government policies are not necessarily exogenous since entrepreneurs can take an active role in affecting these policies (Keim and Hillman, 2008; Pinkse and Groot, 2015). Therefore, because governments have significant influence on the legal and regulatory framework within which entrepreneurs conduct business, entrepreneurs should develop political strategies that constitute an integral part of their overall strategy (Baron, 1995; Hillman and Hitt, 1999; McWilliams et al., 2002).

The political market can potentially provide entrepreneurs with favorable policies, preferential treatment, private information, and access to resources commanded by political actors, which affect entrepreneurs' growth aspirations directly as well as indirectly through their impact in the economic market. By considering the government as another factor of production, entrepreneurs can act politically to "reduce their own production and transaction costs in order to improve their ability to provide cheaper, better, or unique products to theory customers (e.g., through government subsidies), while raising the costs of their rivals at home and abroad (e.g., through government protection)" (Boddewyn and Brewer, 1994, p. 133). Political connections can thus create a strategic advantage by helping a business to defend and maintain market share (Lawton et al., 2013). They can also help reduce environmental uncertainty and transaction costs, and increase long-term sustainability (Lawton et al., 2013). As such, politically-connected entrepreneurs often enjoy preferential treatment by state-owned enterprises such as banks and raw material producers, lighter taxation, preferential treatment when competing for government contracts, and relaxed regulatory oversight or stiffer regulatory oversight of their rivals (Claessens et al., 2008; Faccio, 2006; Fisman, 2001; Haveman et al., 2017). Political connections are therefore a valuable resource that can promote entrepreneurship activities. Moreover, such political capital can be coordinated with entrepreneurs' other resources and capabilities in undergirding their market activities and performance (Baron, 1995, 1999). In the next section, we discuss the specific ways that political connections influence entrepreneurial business reinvestment.

2.3. A politically contingent view of entrepreneurial business reinvestment

By considering entrepreneurs' heterogeneity in terms of their political connections, we are able to recast the relationship between institutions and entrepreneurship. The general attitude in China toward private business and entrepreneurs is highly fickle. Government policies and legal institutions regarding private enterprise have gone through recursive stages of strict prohibition, tolerance, accommodation, and encouragement (Chen and Dickson, 2010; Peng, 2004), making the life of entrepreneurs quite turbulent. While studies suggest that the institutional environment in China is improving (Zhou, 2013, 2014), in some provinces reforms are actually reversing and the institutional environment is becoming harsher rather than better for businesses (Banalieva et al., 2015). Thus, entrepreneurs' perception of the institutional environment should influence their decisions such that a weak institutional environment leads them to focus on firm survival while an improving institutional environment encourages them

to focus on growth (Smallbone and Welter, 2001). However, since relationships with political authorities are important to entrepreneurs in China (Claessens et al., 2008; Peng and Luo, 2000; Zhou, 2013), political connections may alter the relationship between the institutional environment and entrepreneurs' reinvestment in their business. Although we expect a negative relationship between institutional deterioration and entrepreneurial reinvestment, we argue that this relationship changes for politically-connected entrepreneurs since their political connections offer protection from the deteriorating environment in several ways.

In dealing with institutional deterioration, studies have highlighted two key mechanisms undergirding the important role of political connections: 1) easier access to critical resources and opportunities and 2) private protection of property rights (Faccio, 2006; Park et al., 2006; Xin and Pearce, 1996; Zhou, 2013). With the state maintaining control over critical resources (e.g., land, bank loans, and entry permits) and shaping the 'rules of the game' through institutional and administrative actions (Fligstein and Zhang, 2011; Lin, 2011; Peng and Luo, 2000; Walder, 1995), politically-connected entrepreneurs enjoy advantages in gaining access to important resources and opportunities that are critical to business profitability and survival (Li et al., 2008; Park et al., 2006; Zhou, 2009). For instance, politically-connected entrepreneurs in China receive preferential access to credit (Zhou, 2009), tax breaks and government bailouts during tough economic times (Li et al., 2008). In turn, these advantages are likely to encourage entrepreneurs to reinvest in their business despite a deteriorating institutional environment. Here, political connections are a means of creating and sustain a strategic advantage that leads entrepreneurs to reinvest in their business.

Additionally, political connections help entrepreneurs obtain protection of property rights in environments where adequate formal protection is lacking (Peng and Luo, 2000; Zhou, 2013, 2014), thus allowing "entrepreneurs to keep the fruit of their investment from predatory behaviors of the government and its agents" (Zhou, 2013, p. 302). In this way, political connections offer entrepreneurs a sense of security that makes them comfortable reinvesting in their business despite a deteriorating institutional environment. Without such protection, entrepreneurs will be reluctant to reinvest in their business since their private property could be confiscated by the government (North, 1990). Indeed, the private property of entrepreneurs is often expropriated when they are seen as very successful (Smallbone and Welter, 2001). Thus, in a deteriorating institutional environment whereby property rights are weak, political connections are likely necessary for entrepreneurs to reinvest in their business.

Political connections can benefit entrepreneurs in a symbolic way as well. Entrepreneurs' political connections not only serve as pipes that channel resources, opportunities, and private protections, they are also prisms affecting third party's perception of these entrepreneurs (Podolny, 2001). That is, political connections serve as an informational cue that others rely on to make inferences about the underlying quality of entrepreneurs and their business. In China's politicized economy, strong connections with political authorities should enhance entrepreneurs' legitimacy and status in the eyes of potential partners and customers. In a deteriorating institutional environment, political connections are therefore likely to encourage entrepreneurial reinvestment because the entrepreneurs benefit from an enhanced image in the eyes of third parties which should increase their sense of security and desire to grow their business.

Taken together, we see that political connections may endow opportunities and strategic advantages for entrepreneurs in a deteriorating institutional environment, thereby promoting reinvestment in their business. As such, the importance of political connections can be seen as proportional to the government's dominance and intervention in the marketplace and the associated institutional deficiencies. For entrepreneurs in improving institutional environments, the value of political connections tends to decline (Sun et al., 2010; Xia et al., 2009) and therefore, their impact on entrepreneurial reinvestment is likely to be minimal. However, in a deteriorating institutional environment, political connections act as a "security blanket" for entrepreneurs that bolsters their sense of control over uncertain situations and encourages them to accept more risk (De Carolis et al., 2009). Conversely, a lack of political connections in an institutional environment perceived as deteriorating will likely harm entrepreneurs' confidence and resolve thereby lessening their business reinvestment. Institutional deterioration, therefore, is perceived substantially differently (i.e., opportunities vs. threats) by politically-connected entrepreneurs and by those without political connections. Thus, we propose that political connections alter entrepreneurs' reaction to a deteriorating institutional environment. Formally stated:

Hypothesis 2. Political connections moderate the relationship between the institutional environment and entrepreneurs' reinvestment in the business such that for those entrepreneurs without political connections, institutional deterioration will decrease business reinvestment and for those with political connections, institutional deterioration will increase business reinvestment.

3. Methods

3.1. Data

The data used to test the hypotheses come from a survey of Chinese firms in the private sector conducted in 2006 by the Privately Owned Enterprises Research Project Team, which consists of scholars from government bureaus and research institutes. The survey was designed and administered by the All-China Federation of Industry and Commerce, a semi-official organization of private firms that operate at the national, provincial, city, and county/urban district levels. The research team first generated a nationwide random sample of firms using multistage stratified sampling. Following intensive training sessions organized in different locales, research staff visited firms in the sample and, using a questionnaire, conducted a face-to-face structured interview with each entrepreneur. The dataset of this survey is appropriate for testing our hypotheses because it covers a rich set of

questions, including the reinvestment behaviors, multiple firm characteristics, as well as the socioeconomic and political background of the entrepreneurs.

Surveys were completed in 334 counties and urban districts from 31 provinces and metropolitan areas of China. The wide range of regions included in this national survey enables us to assess the impact of institutional variations. The number of valid responses was 3837. In this study, as our key dependent variable is related to financial information, we found a large number of missing values (nearly 50%) in the key variables we are interested in. We examined the nonresponse bias by separating the total sample into two groups and performed *t*-tests regarding major entrepreneur and firm characteristics (Armstrong and Overton, 1977). While younger and female entrepreneurs were more likely to leave nonresponses (cf. Zhou, 2013), no statistically significant differences were found between respondents and nonrespondents on our key variables (i.e., institutional deterioration). We thus excluded cases with missing values on our dependent variable. Following prior research (Cull and Xu, 2005), only firms with positive profits were included in calculating reinvestment; the final working sample consisted of 1289 firms.

3.2. Measures

3.2.1. Entrepreneurial reinvestment

The dependent variable in this study is a firm's reinvestment rate. Following prior studies (Cull and Xu, 2005; Johnson et al., 2002), we measure reinvestment rate as the percentage of after-tax profits earned in 2005 that were reinvested in the business. We use the logarithmic form in our statistical analysis as the distribution of this variable is skewed. In the transformation, a very small number is assigned to zeros (Wooldridge, 2012).

3.2.2. Institutional deterioration

We measure institutional deterioration by assessing nine aspects of the institutional and political environment in China. Specifically, we look at barriers to market access and entry, financing environment, support for science and technology innovation, development of industrial and business associations, construction of enterprise credit system, protection of private property rights and legal rights, government regulation and intervention, policy elucidation and coordination, and the creation of a positive public atmosphere. These elements of the institutional environment have been shown to be critical to entrepreneurship (Aidis et al., 2008; Autio and Fu, 2014; Sobel, 2008), and therefore, entrepreneurs should be responsive to dynamics and movements in them. In China, the institutional environment varies across different regions; firms are thus exposed to different local institutional contexts. It is very difficult to objectively and fully capture the institutional environment facing firms operating in different cities or counties. We therefore measure institutional deterioration in a perceptual way to encompass the local nuances. To capture entrepreneurs' perception of the institutional environment, the respondents were asked to rank these nine aspects along a 4-point scale by considering the dynamics of these institutions in the past year (1 = improved; 4 = worse). A higher score suggests a higher degree of institutional deterioration. Results of factor analysis (see Table 1) suggest that these nine items converge to one single factor, we thus aggregate them to construct a composite index of institutional deterioration (Cronbach's alpha = 0.889).

3.2.3. Political connection

Informed by extant studies (Li et al., 2008; Wang and Qian, 2011), we use the entrepreneur's affiliation with the state's political councils as an important indicator of political connection. The survey asked whether the entrepreneur served as a representative in the National People's Congress (NPC) or Chinese People's Political Consultative Conference (CPPCC) at the national, provincial, city, or county/urban district levels. These two councils are the most important political institutions in which entrepreneurs have opportunities to participate and both can provide political connections to their members (Jia, 2014; Ma and Parish, 2006). Following prior research, this variable has a binary coding (yes = 1).

We control for firm, entrepreneurial, and environmental characteristics. At the firm level, we first include *firm size* and *firm age* in our models. Due to liability of newness and smallness, small firms and young firms should be less capable of exploiting

Table 1
Rotated factor solution for institutional deterioration.

Items	Factor 1 Institutional deterioration
1. Barrier on market access and entry	0.651
2. Financing environment	0.721
3. Support for science and technology innovation	0.762
4. Development of industrial and business associations	0.690
5. Construction of enterprise credit system	0.743
6. Protection of private property rights and legal rights	0.756
7. Government regulation and intervention	0.756
8. Policy elucidation and coordination	0.780
9. Creation of a positive public atmosphere	0.750
Cumulative variance explained	54.06%

Extraction method: principal component analysis.

Rotation method: varimax with Kaiser normalization.

entrepreneurial opportunities, thereby having lower reinvestment rates (Aldrich and Auster, 1986). Firm size is measured as the number of employees of a firm (logged), and firm age is measured as number of years that have passed since the start of the business. As firms may reinvest more when they grow in size, we thus also include a binary variable of *increase in firm size* to indicate whether the firm's employee size increased in the previous year. Resources and financial capabilities could also affect entrepreneurial ability and activities (Alvarez and Busenitz, 2001; Barney, 1991). A firm's *return on sales (ROS)*, as a main index of firm performance, was therefore controlled for and is calculated as net profits over total sales. *Public firms*, with more financing sources, may also be more growth oriented. We thus control for a dummy variable of public firms. Firms located in different places may experience varying institutional and market environments. To control for this potential difference, we include two dummy variables. First, firms *located in a city* are expected to operate within a different environment compared to those in rural places as the general market infrastructures are more advanced in and around cities. Second, in China the government has set up economic development zones that enjoy preferential policies mainly for attracting foreign or nonlocal investments. *Located in economic development zone* thus is expected to present a distinct institutional environment to the firms (Armanios et al., 2016; Hu, 2007). Studies also suggest that a *firm's entertainment fee* is a proxy for the level of corruption in China's context (Cai et al., 2011; Wang, 2013), and the *unauthorized levies* imposed on the firms manifest the state exploitation and political hazards (Jia and Mayer, 2015). As these institutional conditions influence firms' activity, we control for both variables.

Because entrepreneurs' demographic characteristics, human capital, career and social background may explicitly or implicitly impact entrepreneurial activities (Carter et al., 2007; Cooper et al., 1994; Hambrick and Mason, 1984; Packalen, 2007; Shrader and Siegel, 2007; Wright et al., 2007), we control for *entrepreneur age*, *gender*, and *educational background*. Age is measured in number of years. Gender is a dummy variable (1 = male and 0 = female). Education is coded as an ordinal variable ranging from 1 to 6 (1 = "elementary school and below," 2 = "middle school," 3 = "high school," 4 = "junior college," 5 = "university," 6 = "graduate school").

Industry, general market, and economic environment factors are also considered. A series of *industry dummies* are included in all of the models. Local economic and market development, which differ dramatically across Chinese provinces, may also be critical to entrepreneurship. We thus control for two regional variables: *market development* and *GDP growth rate*. Market development is measured by the index of market development of Chinese provinces developed by the National Economic Research Institute (NERI), a widely used statistic (Jia, 2014; Li and Qian, 2013; Shi et al., 2012). GDP growth rate is often used to capture local economic development. We therefore use the lagged provincial GDP growth rate, which is measured as the ratio of GDP in 2005 over GDP in 2004 (minus one).

3.3. Models

We used ordinal least squares (OLS) regression models and Huber-White's robust standard errors to correct for nonspherical disturbances. To check for multicollinearity between independent and dependent variables, we examined the Variance Inflation Factor (VIF). The VIFs for the independent variables ranged from 1.03 to 4.71 with a mean VIF of 1.58, which is substantially less than the "rule of thumb" of 10, indicating no serious multicollinearity problem with the data (Greene, 2011). To assess the potential for endogeneity bias, we utilized instrumental variables for both institutional deterioration and political connection (Bascle, 2008; Hamilton and Nickerson, 2003). Entrepreneurs' party background and engagement in philanthropy are potentially related to entrepreneurs' political connections and their perceptions of institutional environment (Ma and Parish, 2006; Wang and Qian, 2011), and there are no clear and direct connections between these variables and the firm's willingness to reinvest. Following practices in the literature (Knockaert et al., 2015; Kreutzer et al., 2015), we first performed the two-stage least-squares regression and then used the Durbin-Wu-Hausman (DWH) test to examine endogeneity. When utilizing corporate philanthropy (i.e., a dummy variable capturing whether the entrepreneur engaged in the *guangcai* philanthropic project in China) and an entrepreneur's party background (i.e., a dummy variable measuring whether the entrepreneur is a member of non-Communist parties) as instruments, the resulting DWH tests for institutional deterioration ($F = 0.60$, $p = 0.44$; $\chi^2 = 0.61$, $p = 0.44$) and political connection ($F = 1.26$, $p = 0.26$; $\chi^2 = 0.62$, $p = 0.46$) showed that the independent variables in question can be considered exogenous. The OLS estimated are thus unbiased, and we accordingly report the OLS regression results in the following section.

4. Results

The descriptive statistics and correlations are shown in Table 2 and Table 3. Table 4 presents the results of regression models testing the moderating role of political connection on the relationship between institutional deterioration and entrepreneurial reinvestment. Model 1 includes the control variables, Model 2 shows the main effects, and Model 3 reports the moderation effect. While multiple control variables were significant, particularly the negative effects of education and market development deserve a brief discussion. Education is often negatively related to entrepreneurial behavior and venture performance (e.g., Chrisman et al., 2005), indeed a meta-analysis for developing countries showed that more highly educated workers gravitate toward wage jobs and tend to avoid entrepreneurial behavior (van der Sluis et al., 2005). Furthermore, it is possible that individuals with higher education may recognize the need to diversify their wealth portfolio and extract money from the business for other investments, particularly considering the potentially volatile institutional environment in China that they operate in. Similarly, the provinces with a higher level of market development are often the richer economies in China, where entrepreneurs will have most investment options outside their respective businesses. Furthermore, in these provinces, market competition is fiercer, which may further reduce the reinvestment rate.

Model 2 shows that the main effect of institutional deterioration on entrepreneurial reinvestment is not significant. However, the product item of institutional deterioration and political connection shown in Model 3 is significant ($b = 1.391, p < 0.01$), suggesting a moderating effect of political connection. To better understand and interpret the results, based on the coefficients in Model 3 we plot the moderation in Fig. 1. As we can see, there is a negative relationship between institutional deterioration and reinvestment rate for nonpolitically-connected entrepreneurs. In contrast, institutional deterioration positively affects reinvestment rate for politically-connected entrepreneurs. The results, therefore, lend strong support for our hypothesis of the politically contingent effect of institutional deterioration on entrepreneurial reinvestment.

To assess the practical impact of the moderating role of political connections, we calculated the difference of reinvestment rate between politically-connected entrepreneurs and nonpolitically-connected entrepreneurs. As shown in Fig. 2, as the level of institutional deterioration increases, the difference in reinvestment rates becomes larger. In our sample, the median of institutional deterioration is 2.0, and the standard deviation (SD) is 0.49. When the level of institutional deterioration is 2.0, the difference of reinvestment rate caused by political connections is 1.77%; a SD larger; when deterioration increases to 2.5, the gap becomes 6.26%, which is a significant difference. As the deterioration approaches 3, the gap of reinvestment rate goes beyond 11%. These differences induced by political connections demonstrate the strong practical significance of our findings.

4.1. Robustness tests

In a further step, we split the whole sample into two subsamples and separately examine the relationship between institutional deterioration and reinvestment rate of nonpolitically-connected entrepreneurs versus politically-connected entrepreneurs. As shown in the left panel of Table 5, institutional deterioration has a significant and negative effect on reinvestment rate ($b = -0.674, p < 0.05$) for nonpolitically-connected entrepreneurs. For politically-connected entrepreneurs, the impact of institutional deterioration on reinvestment rate becomes positive and significant ($b = 0.721, p < 0.05$). This striking contrast further lends support to our hypothesis that entrepreneurs' political connections moderate the relationship between institutional deterioration and entrepreneurial reinvestment. Thus, entrepreneurs' interpretations and reaction to institutional deterioration are likely contingent on their political connections that endow advantages in the political market.

Several additional analyses corroborate the robustness of our findings. We control for the region-level variables of market and economic development to consider other possible impacts of the regional environment. In running models including a set of province dummies, our findings remained unchanged. Moreover, as entrepreneurs who are *former cadres* may also have opportunities to develop political connections (G. Fan et al., 2007; J.P.H. Fan et al., 2007; Li et al., 2008), we constructed an alternative measure of political connections. Specifically, we considered an entrepreneur as politically-connected if he/she is a prior cadre of the Communist Party of China or the government. When we included the prior cadres as political entrepreneurs, the results were consistent with our main findings.

Table 2
Measures and descriptive statistics.

Variables	Measures	Mean	SD
Reinvestment rate	Percentage of after-tax profits earned in 2005 that were reinvested in the business	0.47	0.35
Institutional deterioration	Average score of nine items of the institutional and political environment in China on 4-point scale	2.02	0.49
Entrepreneur's political connection	1 = if the entrepreneur served as a representative in the political councils including PC and CPPCC; 0 otherwise	0.38	0.49
Firm size	The number of employees of a firm (logged)	3.97	1.60
Increase of firm size	1 = if the firm increases in its employee size; 0 otherwise	0.47	0.50
Firm age	The number of years that have passed since the start of the business	7.02	4.38
ROS	Net profits over total sales	0.09	0.18
Public firm	1 = if the firm is a publically traded company; 0 otherwise	0.02	0.14
Located in city	1 = if the firm is located in a city; 0 otherwise	0.50	0.50
Located in economic development zone	1 = if the firm is located in a economic development zone (<i>Kaifā Qū</i>)	0.10	0.30
Entertainment fee	Expenditures of entertainment and public relations, scaled by sales	0.76	2.72
Unauthorized levies	The amount of unauthorized levies (<i>Tan Pai</i>) the firm paid in 2005 (logged)	1.27	0.93
Entrepreneur age	The number of years	44.40	8.34
Entrepreneur gender	1 = male, 0 = female	0.86	0.35
Entrepreneur education	1 = "elementary school and below," 2 = "middle school," 3 = "high school," 4 = "junior college," 5 = "university," 6 = "graduate school"	3.54	1.07
Industries	Twelve dummy variables capturing thirteen industries		
Market development	The index of market development of Chinese provinces developed by the National Economic Research Institute	8.20	1.83
GDP growth rate	The ratio of GDP in 2005 over GDP in 2004 (minus one)	0.17	0.04
Former cadre	1 = if the entrepreneur was a cadre in the party or government at or above the county and division level (<i>Xian Chu Ji</i>); 0 otherwise	0.18	0.38

Notes: the industries include agriculture, mining, manufacturing, utility, construction, transportation, IT, retail, restaurant, real estate, education, health care, and others.

Table 3

Variables and correlations.

Variables	1	2	3	4	5	6	7	8	9	10	11	12	13	14	15	16
1. Reinvestment rate																
2. Institutional deterioration	−0.010															
3. Political connection	0.108	0.007														
4. Firm size	0.102	0.021	0.420													
5. Increase of firm size	0.123	−0.028	0.167	0.322												
6. Firm age	0.118	0.048	0.276	0.257	0.115											
7. ROS	0.125	−0.020	−0.015	−0.033	0.038	0.024										
8. Public firm	0.034	−0.056	0.051	0.041	−0.013	0.013	0.023									
9. Located in city	−0.037	0.046	0.025	−0.050	−0.045	0.020	0.006	−0.007								
10. Located in development zone	0.019	−0.009	0.005	0.050	0.045	−0.001	0.009	−0.002	−0.336							
11. Entertainment fee	−0.082	0.006	−0.013	−0.079	0.005	−0.007	−0.232	0.003	0.031	−0.007						
12. Unauthorized levies	0.319	0.000	0.097	0.190	0.037	0.092	0.053	0.079	0.037	−0.004	−0.047					
13. Entrepreneur age	−0.009	0.018	0.202	0.187	0.013	0.233	−0.032	−0.021	−0.038	−0.007	−0.036	0.031				
14. Entrepreneur gender	0.036	−0.014	0.068	0.120	0.017	0.043	−0.021	−0.025	−0.075	0.028	−0.001	0.017	0.110			
15. Entrepreneur education	−0.078	0.031	0.112	0.179	0.061	−0.021	−0.001	−0.001	0.171	0.068	−0.009	0.071	−0.141	−0.003		
16. Market development	−0.093	0.027	−0.141	−0.001	−0.032	0.057	−0.039	0.000	−0.120	0.036	−0.010	0.021	0.007	−0.001	−0.042	
17. GDP growth rate	0.062	0.013	0.081	0.115	0.053	0.020	0.002	−0.021	−0.049	−0.045	0.011	−0.061	0.076	0.032	−0.013	−0.014

Notes: $N = 1289$; correlations >0.055 are significant at the level of $p < 0.05$.

Table 4

Results of regressions models on entrepreneurial reinvestment rate.

Variables	Reinvestment rate					
	Model 1		Model 2		Model 3	
Firm size	−0.039	(0.085)	−0.071	(0.090)	−0.060	(0.090)
Increase of firm size	0.631**	(0.234)	0.611*	(0.237)	0.604*	(0.237)
Firm age	0.103***	(0.029)	0.102***	(0.030)	0.101***	(0.030)
ROS	2.566***	(0.536)	2.838***	(0.549)	2.815***	(0.548)
Public firm	−0.202	(0.979)	0.024	(1.055)	−0.073	(1.053)
Located in city	−0.145	(0.253)	−0.019	(0.256)	−0.048	(0.255)
Located in development zone	0.221	(0.407)	0.339	(0.408)	0.355	(0.407)
Entertainment fee	−0.098*	(0.040)	−0.074	(0.048)	−0.074	(0.048)
Unauthorized levies	1.342***	(0.120)	1.329***	(0.122)	1.327***	(0.121)
Entrepreneur age	−0.037*	(0.015)	−0.041**	(0.015)	−0.042**	(0.015)
Entrepreneur gender	0.501	(0.355)	0.625†	(0.358)	0.611†	(0.357)
Entrepreneur education	−0.355**	(0.113)	−0.424***	(0.116)	−0.426***	(0.116)
Industry dummies	Y		Y		Y	
Market development	−0.244***	(0.064)	−0.211**	(0.066)	−0.213**	(0.065)
GDP growth rate	6.067*	(2.531)	5.622*	(2.544)	5.832*	(2.538)
Institutional deterioration			−0.110	(0.242)	−0.663*	(0.310)
Entrepreneur political connection			0.346	(0.263)	−2.488*	(1.031)
Institutional deterioration × Political connection					1.391**	(0.490)
Constant	−2.093†	(1.188)	−1.667	(1.281)	−0.492	(1.343)
N	1289		1289		1289	
R ²	0.175		0.180		0.186	

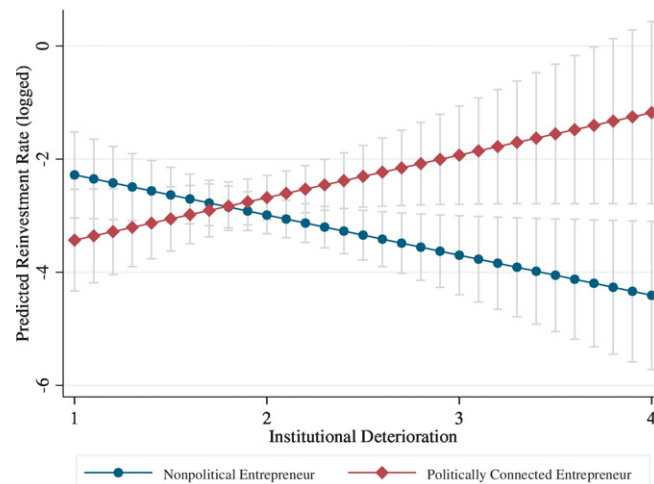
Notes: standard errors in parentheses; twelve dummy variables indicating industries are included in all models.

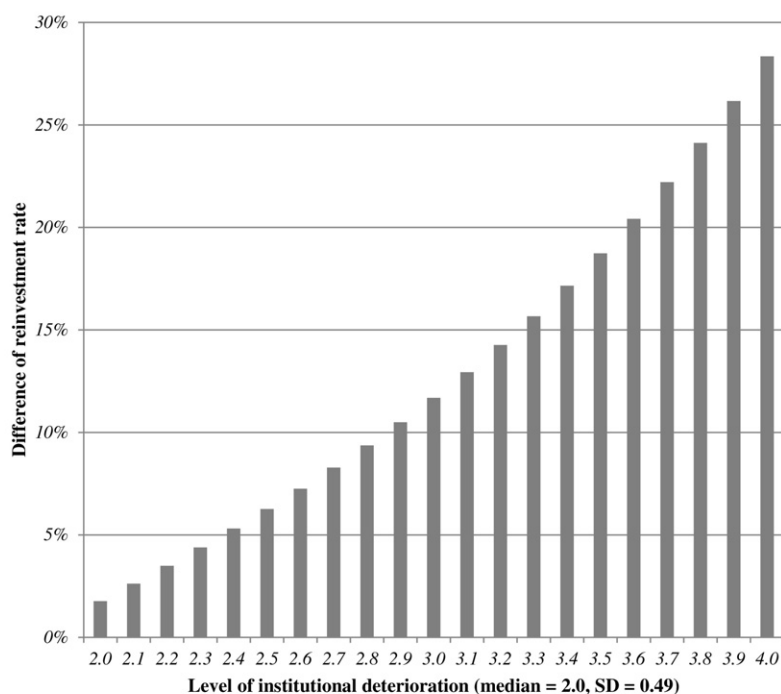
† $p < 0.1$.* $p < 0.05$.** $p < 0.01$.*** $p < 0.001$.

5. Discussion and conclusion

We propose a politically contingent view of the relationship between institutions and entrepreneurship by drawing on insights from the political market and political connections literature. Our empirical work on China's entrepreneurs largely supports this proposition by showing that the impact of institutional deterioration on entrepreneurial reinvestment substantially hinges on entrepreneurs' political connections. That is, institutional deterioration is not universally perceived as a threat that undermines entrepreneurial reinvestment; rather, when entrepreneurs have political connections, the advantages endowed encourage entrepreneurs' to reinvest in their business.

Our framework suggests that in an environment perceived as harsh to business, political connections encourage entrepreneurs to see opportunities for growth. In contrast, entrepreneurs that lack political connections will mainly perceive threats in a deteriorating institutional environment and thus, limit their business reinvestment. Our findings therefore highlight how political connections can play a pivotal role in shaping entrepreneurs' view of the institutional environment in a transitioning economy and therefore affect how they respond to their institutional environment.

**Fig. 1.** Contingent impact of institutional deterioration on reinvestment rate.



Note: The difference of reinvestment rate is the reinvestment rate of politically-connected entrepreneurs minus the reinvestment rate of nonpolitical entrepreneurs, holding other variables constant at the mean.

Fig. 2. Difference of reinvestment rate between politically-connected entrepreneurs and nonpolitical entrepreneurs.

This politically contingent relationship between institutions and entrepreneurial reinvestment has both theoretical and empirical implications. First, prior research has mostly focused on the relationship between institutions and entrepreneurship at the country level, drawing on global data. Using aggregated information, however, largely downplays the heterogeneity among individual entrepreneurs. We thus risk making the “ecological fallacy” if we indiscriminately apply these findings to the entrepreneurial level. As shown, the general positive/negative relationship between institutions and entrepreneurship, though perhaps true at

Table 5

Regressions models on entrepreneurial reinvestment rate by political connection.

Variables	Reinvestment rate	
	Non-political entrepreneurs	Politically-connected entrepreneurs
Institutional deterioration	−0.674* (0.328)	0.721* (0.351)
Firm size	0.033 (0.126)	−0.095 (0.127)
Increase of firm size	0.861** (0.326)	0.016 (0.342)
Firm age	0.074† (0.043)	0.093* (0.037)
ROS	2.171*** (0.633)	21.653 (1.123)
Public firm	−1.999 (1.636)	1.578 (1.331)
Located in city	−0.448 (0.353)	0.487 (0.367)
Located in development zone	0.578 (0.571)	−0.120 (0.579)
Entertainment fee	−0.092† (0.054)	0.167 (0.137)
Unauthorized levies	1.468*** (0.172)	1.194*** (0.169)
Entrepreneur age	−0.058** (0.020)	−0.022 (0.023)
Entrepreneur gender	0.514 (0.494)	0.769 (0.520)
Entrepreneur education	−0.422* (0.166)	−0.474** (0.158)
Industry dummies	Y	Y
Market development	−0.359*** (0.096)	−0.023 (0.093)
GDP growth rate	4.200 (3.565)	5.709 (3.575)
Constant	0.964 (1.773)	−3.927* (1.903)
N	749	540
R ²	0.202	0.207

Notes: standard errors in parentheses; twelve dummy variables indicating industries are included in all models.

† $p < 0.1$.

* $p < 0.05$.

** $p < 0.01$.

*** $p < 0.001$.

the country level, does not hold for individual entrepreneurs because important intra-country and inter-entrepreneur heterogeneity have been ignored. Future research analyzing entrepreneurs in transitioning economies should therefore take a more fine-grained approach, recognizing variance among entrepreneurs and the perceived institutional environment.

Second, we draw attention to entrepreneurs' political connections and suggest that such political capital can shape entrepreneurs' interpretation and reaction to their institutional environment, thereby affecting their decisions. Echoing recent developments in the field of strategy showing that political connections have important economic and business implications (Lawton et al., 2013; Lux et al., 2011), we highlight the need to study how connections can be leveraged by entrepreneurs in a transitioning economy. Though various fields have long suggested that business and politics are highly intertwined, research has not yet paid sufficient attention to the political side of entrepreneurship. Yet, entrepreneurs, due to their heterogeneity in political capital, are stratified in the political market where their relative advantages or disadvantages will be ultimately channeled into the competitive market. Thus, entrepreneurs' decisions and activities are not solely determined by their opportunities in the economic and product markets, but also hinge on their political power and the resources gained from political connections. Future research should build on our study by examining the role of political connections in encouraging entrepreneurs in both transitioning and advanced economies to reinvest in their business and pursue growth strategies.

Third, in considering the relationship between institutions and entrepreneurship, we should not assume that institutions are exogenous to entrepreneurs. The contingent impact of institutions on entrepreneurship implies that entrepreneurs are, to varying degrees, susceptible to institutional influences. Moreover, because entrepreneurs are political actors whose strength of political connections vary (Pinkse and Groot, 2015), some may exploit those connections in a weak institutional environment for private interest. Even in the West, entrepreneurs and other business actors have varying connections with politicians, political parties, NGOs, and social movement organizations and also possess varying levels of power to influence the institutional environment in which they are embedded. The relationship between the institutional environment and entrepreneurial outcomes is expected to be contingent on these connections as well in these contexts, which future research should examine.

5.1. Limitations and future research

We acknowledge several research limitations. While China is particularly appropriate for situating and testing our key argument and hypothesis (Puffer et al., 2010), the generalizability of our results is limited by the single-country context; our study should thus be replicated in other transitioning and emerging economies as well as advanced economies. Indeed, the importance of political connections has been revealed in other transitioning and emerging economies (Claessens et al., 2008; Puffer et al., 2010) and in developed countries, entrepreneurs have varying connections with politicians and political parties that could affect their strategic decisions and performance. The relationship between the perceived institutional environment and entrepreneurial outcomes may be contingent on these connections as well in these contexts, which future research should examine. Moreover, China's political market and environment are intriguing but somewhat distinct (Lin, 2001). To more comprehensively capture the political dynamics of entrepreneurship, future research should utilize cross-country and comparative institutional approaches to better understand the role of politics in entrepreneurship. Further, we need to acknowledge how some research has tied political connections to corruption (e.g., Lawton et al., 2013; Puffer et al., 2010). While it was beyond the scope of our study to explore if political connections lead to an unfair advantage, future research may want to explore the business outcomes associated with political connections in a corrupt setting. Our research suggests that political connections lead to higher levels of business reinvestment when entrepreneurs perceive a deteriorating institutional environment. Future research, via longitudinal designs, should validate if the reinvestment actually yields financial benefits for the entrepreneur and if those political connections have a less significant effect as the institutional environment improves.

Lastly, we need to acknowledge the nature of our sample. Consistent with the setting of our study, the entrepreneurs in our sample were primarily founders of SMEs. While the average business age in our sample is 7.02 years old, it falls within operationalizations of recent research in the field of entrepreneurship (e.g., Wasserman (2016, mean firm age = 6.93), Thorgren et al. (2016, mean firm age = 9.7), Plummer et al. (2016, mean firm age = 7.45)). It would therefore be interesting to replicate our study with nascent ventures.

In conclusion, our study highlights the importance of political connections to entrepreneurs in a deteriorating institutional environment. Drawing attention to how the political arena can create and sustain advantages for entrepreneurs, we argue that political connections affect how entrepreneurs react to a deteriorating institutional environment. While those that lack political connections will limit reinvestment, those with political connections will increase the reinvestment in their business. Further, as the perceived deterioration of the institutional environment increases, political connections have a greater positive affect on entrepreneurial reinvestment. Our study therefore supports a politically contingent view of the impact of institutional deterioration on entrepreneurs' reinvestment in their business.

Acknowledgment

We are grateful to the editor Roy Suddaby and three anonymous reviewers for their insightful and constructive comments and suggestions. We also appreciate helpful suggestions we received from Wubiao Zhou on an earlier version of this paper. The first author acknowledges the support from National Natural Science Foundation of China (No. 71602185).

References

- Aidis, R., Estrin, S., Mickiewicz, T., 2008. Institutions and entrepreneurship development in Russia: a comparative perspective. *J. Bus. Ventur.* 23, 656–672.
- Aldrich, H.E., Auster, E.R., 1986. Even dwarfs started small: liabilities of age and size and their strategic implications. *Res. Organ. Behav.* 8, 165–198.
- Alvarez, S.A., Busenitz, L.W., 2001. The entrepreneurship of resource-based theory. *J. Manag.* 27, 755–775.
- Armanios, D.E., Eesley, C.E., Li, J., Eisenhardt, K.M., 2016. How entrepreneurs leverage institutional intermediaries in emerging economies to acquire public resources. *Strateg. Manag. J.* (Forthcoming).
- Armstrong, J.S., Overton, T.S., 1977. Estimating nonresponse bias in mail surveys. *J. Mark. Res.* 14, 396–402.
- Autio, E., Fu, K., 2014. Economic and political institutions and entry into formal and informal entrepreneurship. *Asia Pac. J. Manag.* 1–28.
- Banalieva, E.R., Eddleston, K.A., Zellweger, T.M., 2015. When do family firms have an advantage in transitioning economies? Toward a dynamic institution-based view. *Strateg. Manag. J.* 36, 1358–1377.
- Barney, J., 1991. Firm resources and sustained competitive advantage. *J. Manag.* 17, 99–120.
- Baron, D.P., 1995. Integrated strategy: market and nonmarket components. *Calif. Manag. Rev.* 37, 47–65.
- Baron, D.P., 1999. Integrated market and nonmarket strategies in client and interest group politics. *Bus. Polit.* 1, 7–34.
- Bascle, G., 2008. Controlling for endogeneity with instrumental variables in strategic management research. *Strateg. Organ.* 6, 285–327.
- Batjargal, B., Hitt, M.A., Tsui, A.S., Arregle, J.-L., Webb, J.W., Miller, T.L., 2013. Institutional polycentrism, entrepreneurs' social networks, and new venture growth. *Acad. Manag. J.* 56, 1024–1049.
- Boddeyn, J.J., Brewer, T.L., 1994. International-business political behavior: new theoretical directions. *Acad. Manag. Rev.* 19, 119–143.
- Bonardi, J.-P., Hillman, A.J., Keim, G.D., 2005. The attractiveness of political markets: implications for firm strategy. *Acad. Manag. Rev.* 30, 397–413.
- Bruton, G.D., Ahlstrom, D., Li, H.-L., 2010. Institutional theory and entrepreneurship: where are we now and where do we need to move in the future? *Entrep. Theory Pract.* 34, 421–440.
- Bruton, G.D., Ketchen Jr., D.J., Ireland, R.D., 2013. Entrepreneurship as a solution to poverty. *J. Bus. Ventur.* 28, 683–689.
- Buchanan, J.M., 1987. The constitution of economic policy. *Am. Econ. Rev.* 77, 243–250.
- Cai, H., Fang, H., Xu, L.C., 2011. Eat, drink, firms, government: an investigation of corruption from the entertainment and travel costs of Chinese firms. *J. Law Econ.* 54, 55–78.
- Carter, S., Shaw, E., Lam, W., Wilson, F., 2007. Gender, entrepreneurship, and bank lending: the criteria and processes used by bank loan officers in assessing applications. *Entrep. Theory Pract.* 31, 427–444.
- Chen, J., Dickson, B.J., 2010. *Allies of the State: China's Private Entrepreneurs and Democratic Change*. Harvard University Press, Cambridge, MA.
- Chen, C.J.P., Li, Z., Su, X., Sun, Z., 2011. Rent-seeking incentives, corporate political connections, and the control structure of private firms: Chinese evidence. *J. Corp. Finan.* 17, 229–243.
- Chrisman, J.J., McMullan, E., Hall, J., 2005. The influence of guided preparation on the long-term performance of new ventures. *J. Bus. Ventur.* 20, 769–791.
- Claessens, S., Feijen, E., Laeven, L., 2008. Political connections and preferential access to finance: the role of campaign contributions. *J. Financ. Econ.* 88, 554–580.
- Cooper, A.C., Gimeno-Gascon, F.J., Woo, C.Y., 1994. Initial human and financial capital as predictors of new venture performance. *J. Bus. Ventur.* 9, 371–395.
- Cuervo-Cazurra, A., Dau, L.A., 2009. Promarket reforms and firm profitability in developing countries. *Acad. Manag. J.* 52, 1348–1368.
- Cull, R., Xu, L.C., 2005. Institutions, ownership, and finance: the determinants of profit reinvestment among Chinese firms. *J. Financ. Econ.* 77, 117–146.
- De Carolis, D.M., Litzky, B.E., Eddleston, K.A., 2009. Why networks enhance the progress of new venture creation: the influence of social capital and cognition. *Entrep. Theory Pract.* 33, 527–545.
- Ebner, J., 2014. *Entrepreneurs: China's Next Growth Engine?* The Diplomat.
- Estrin, S., Korosteleva, J., Mickiewicz, T., 2013. Which institutions encourage entrepreneurial growth aspirations? *J. Bus. Ventur.* 28, 564–580.
- Eunni, R.V., Manolova, T.S., 2012. Are the BRIC economies entrepreneur-friendly? An institutional perspective. *J. Ent. Cult.* 20, 171–202.
- Faccio, M., 2006. Politically-connected firms. *Am. Econ. Rev.* 96, 369–386.
- Fan, G., Wang, X., Zhu, H., 2007a. NERI Index of Marketization of China's Provinces. Economics Science Press, Beijing.
- Fan, J.P.H., Wong, T.J., Zhang, T., 2007b. Politically-connected CEOs, corporate governance, and Post-IPO performance of China's newly partially privatized firms. *J. Financ. Econ.* 84, 330–357.
- Fisman, R., 2001. Estimating the value of political connections. *Am. Econ. Rev.* 91, 1095–1102.
- Fligstein, N., Zhang, J., 2011. A new agenda for research on the trajectory of Chinese capitalism. *Manag. Organ. Rev.* 7, 39–62.
- Fogel, K., Hawk, A., Morck, R., Yeung, B., 2006. Institutional obstacles to entrepreneurship. In: Casson, M., Yeung, B., Basu, A., Wadeson, N. (Eds.), *Oxford Handbook of Entrepreneurship*. Oxford University Press, Oxford, pp. 540–579.
- George, G., Rao-Nicholson, R., Corbishley, C., Bansal, R., 2015. Institutional entrepreneurship, governance, and poverty: insights from emergency medical response services in India. *Asia Pac. J. Manag.* 32, 39–65.
- Greene, W.H., 2011. *Econometric Analysis*. seventh ed. Prentice Hall.
- Hambrick, D.C., Mason, P.A., 1984. Upper echelons: the organization as a reflection of its top managers. *Acad. Manag. Rev.* 9, 193–206.
- Hamilton, B.H., Nickerson, J.A., 2003. Correcting for endogeneity in strategic management research. *Strateg. Organ.* 1, 51–78.
- Haveman, H.A., Jia, N., Shi, J., Wang, Y., 2017. The dynamics of political embeddedness in China. *Adm. Sci. Q.* 62, 67–104.
- Hillman, A.J., Hitt, M.A., 1999. Corporate political strategy formulation: a model of approach, participation, and strategy decisions. *Acad. Manag. Rev.* 24, 825–842.
- Hillman, A.J., Keim, G.D., Schuler, D., 2004. Corporate political activity: a review and research agenda. *J. Manag.* 30, 837–857.
- Hu, A.G., 2007. Technology parks and regional economic growth in China. *Res. Policy* 36, 76–87.
- Huang, Y., 2010. *Entrepreneurship in China*. World Financ. Rev.
- Hwang, H., Powell, W.W., 2005. Institutions and entrepreneurship. In: Alvarez, S.A., Agarwal, R., Sorenson, O. (Eds.), *Handbook of Entrepreneurship Research, International Handbook Series on Entrepreneurship*. Springer, pp. 201–232.
- Jennings, P.D., Greenwood, R., Lounsbury, M.D., Suddaby, R., 2013. Institutions, entrepreneurs, and communities: a special issue on entrepreneurship. *J. Bus. Ventur.* 28, 1–9.
- Jia, N., 2014. Are collective political actions and private political actions substitutes or complements? Empirical evidence from China's private sector. *Strateg. Manag. J.* 35, 292–315.
- Jia, N., Mayer, K.J., 2015. Political hazards and firms' geographic concentration. *Strateg. Manag. J.* (Forthcoming).
- Johnson, S., McMillan, J., Woodruff, C., 2002. Property rights and finance. *Am. Econ. Rev.* 92, 1335–1356.
- Keim, G.D., Hillman, A.J., 2008. Political environments and business strategy: implications for managers. *Bus. Horiz.* 51, 47–53.
- Knockaert, M., Bjornali, E.S., Erikson, T., 2015. Joining forces: top management team and board chair characteristics as antecedents of board service involvement. *J. Bus. Ventur.* 30, 420–435.
- Kreutzer, M., Walter, J., Cardinal, L.B., 2015. Organizational control as antidote to politics in the pursuit of strategic initiatives. *Strateg. Manag. J.* 36, 1317–1337.
- Kriauciunas, A., Kale, P., 2006. The impact of socialist imprinting and search on resource change: a study of firms in Lithuania. *Strateg. Manag. J.* 27, 659–679.
- Kshetri, N., 2007. Institutional changes affecting entrepreneurship in China. *J. Dev. Entrep.* 12, 415–432.
- Lawton, T., McGuire, S., Rajwani, T., 2013. Corporate political activity: a literature review and research agenda. *Int. J. Manag. Rev.* 15, 86–105.
- Li, J., Qian, C., 2013. Principal-principal conflicts under weak institutions: a study of corporate takeovers in China. *Strateg. Manag. J.* 34, 498–508.
- Li, H., Meng, L., Wang, Q., Zhou, L., 2008. Political connections, financing and firm performance: evidence from Chinese private firms. *J. Dev. Econ.* 87, 283–299.
- Lin, Y., 2001. *Between Politics and Markets: Firms, Competition, and Institutional Change in Post-Mao China*. Cambridge University Press, New York, NY.
- Lin, N., 2011. Capitalism in China: a centrally managed capitalism (CMC) and its future. *Manag. Organ. Rev.* 7, 63–96.
- Loyalka, M.D., 2006. A Chinese Welcome for Entrepreneurs. *Business Week Online* (January 6).
- Lux, S., Crook, T.R., Woehr, D.J., 2011. Mixing business with politics: a meta-analysis of the antecedents and outcomes of corporate political activity. *J. Manag.* 37, 223–247.

- Ma, D., Parish, W.L., 2006. Tocquevillian moments: charitable contributions by Chinese private entrepreneurs. *Soc. Forces* 85, 943–964.
- McCarthy, A.M., Schoorman, F.D., Cooper, A.C., 1993. Reinvestment decisions by entrepreneurs: rational decision-making or escalation of commitment? *J. Bus. Ventur.* 8, 9–24.
- McMillan, J., Woodruff, C., 1999. Dispute prevention without courts in Vietnam. *J. Law Econ. Org.* 15, 637–658.
- McMillan, J., Woodruff, C., 2002. The central role of entrepreneurs in transition economies. *J. Econ. Perspect.* 16, 153–170.
- McMullen, J.S., Bagby, D.R., Palich, L.E., 2008. Economic freedom and the motivation to engage in entrepreneurial action. *Entrep. Theory Pract.* 32, 875–895.
- McWilliams, A., Van Fleet, D.D., Cory, K.D., 2002. Raising rivals' costs through political strategy: an extension of resource-based theory. *J. Manag. Stud.* 39, 707–724.
- Meyer, K.E., Estrin, S., Bhaumik, S.K., Peng, M.W., 2009. Institutions, resources, and entry strategies in emerging economies. *Strateg. Manag. J.* 30, 61–80.
- Ministry of Commerce People's Republic of China, 2012. <http://english.mofcom.gov.cn/> Accessed, 04/26/2017.
- North, D.C., 1990. *Institutions, Institutional Change and Economic Performance*. Cambridge University Press, New York, NY.
- Oliver, C., Holzinger, I., 2008. The effectiveness of strategic political management: a dynamic capabilities framework. *Acad. Manag. Rev.* 33, 496–520.
- Packalen, K.A., 2007. Complementing capital: the role of status, demographic features, and social capital in founding teams' abilities to obtain resources. *Entrep. Theory Pract.* 31, 873–891.
- Park, S.H., Li, S., Tse, D.K., 2006. Market liberalization and firm performance during China's economic transition. *J. Int. Bus. Stud.* 37, 127–147.
- Peng, Y., 2004. Kinship networks and entrepreneurs in China's transitional economy. *Am. J. Sociol.* 109, 1045–1074.
- Peng, M.W., Jiang, Y., 2010. Institutions behind family ownership and control in large firms. *J. Manag. Stud.* 47, 253–273.
- Peng, M.W., Luo, Y., 2000. Managerial ties and firm performance in a transition economy: the nature of a micro-macro link. *Acad. Manag. J.* 43, 486–501.
- Pinkse, J., Groot, K., 2015. Sustainable entrepreneurship and corporate political activity: overcoming market barriers in the clean energy sector. *Entrep. Theory Pract.* 39, 633–654.
- Pistru, D., Huang, W., Oksoy, D., Jing, Z., Welsch, H., 2001. Entrepreneurship in China: characteristics, attributes, and family forces shaping the emerging private sector. *Fam. Bus. Rev.* 14, 141–152.
- Plummer, L.A., Allison, T.H., Connelly, B.L., 2016. Better together? Signaling interactions in new venture pursuit of initial external capital. *Acad. Manag. J.* 59, 1585–1604.
- Podolny, J.M., 2001. Networks as the pipes and prisms of the market. *Am. J. Sociol.* 107, 33–60.
- Powell, W.W., 1991. Expanding the scope of institutional analysis. In: Powell, W.W., DiMaggio, P.J. (Eds.), *The New Institutionalism in Organizational Analysis*. University of Chicago Press, Chicago, pp. 183–203.
- Puffer, S.M., McCarthy, D.J., Boisot, M., 2010. Entrepreneurship in Russia and China: the impact of formal institutional voids. *Entrep. Theory Pract.* 34, 441–467.
- Scott, W.R., 2014. *Institutions and Organizations: Ideas, Interests and Identities*. fourth ed. SAGE Publications, Thousand Oaks, CA.
- Shi, W. (Stone), Sun, S.L., Peng, M.W., 2012. Sub-national institutional contingencies, network positions, and IJV partner selection. *J. Manag. Stud.* 49, 1221–1245.
- Shrader, R., Siegel, D.S., 2007. Assessing the relationship between human capital and firm performance: evidence from technology-based new ventures. *Entrep. Theory Pract.* 31, 893–908.
- Siegel, J., 2007. Contingent political capital and international alliances: evidence from south Korea. *Adm. Sci. Q.* 52, 621–666.
- van der Sluis, J., van Praag, M., Vijverberg, W., 2005. Entrepreneurship selection and performance: a meta-analysis of the impact of education in developing economies. *World Bank Econ. Rev.* 19, 225–261.
- Smallbone, D., Welter, F., 2001. The distinctiveness of entrepreneurship in transition economies. *Small Bus. Econ.* 16, 249–262.
- Sobel, R.S., 2008. Testing Baumol: institutional quality and the productivity of entrepreneurship. *J. Bus. Ventur.* 23, 641–655.
- Stenholm, P., Acs, Z.J., Wuebker, R., 2013. Exploring country-level institutional arrangements on the rate and type of entrepreneurial activity. *J. Bus. Ventur.* 28, 176–193.
- Sun, P., Mellahi, K., Thun, E., 2010. The dynamic value of MNE political embeddedness: the case of the Chinese automobile industry. *J. Int. Bus. Stud.* 41, 1161–1182.
- Thorgren, S., Sirén, C., Nordström, C., Wincent, J., 2016. Hybrid entrepreneurs' second-step choice: the nonlinear relationship between age and intention to enter full-time entrepreneurship. *J. Bus. Ventur.* 5, 14–18 (Insights).
- Valdez, M.E., Richardson, J., 2013. Institutional determinants of macro-level entrepreneurship. *Entrep. Theory Pract.* 37, 1149–1175.
- Walder, A.G., 1995. Local governments as industrial firms: an organizational analysis of China's transitional economy. *Am. J. Sociol.* 101, 263–301.
- Wang, Y., 2013. Institutions and bribery in an authoritarian state. *Stud. Comp. Int. Dev.* 49, 217–241.
- Wang, H., Qian, C., 2011. Corporate philanthropy and financial performance of Chinese firms: the roles of social expectations and political access. *Acad. Manag. J.* 54, 1159–1181.
- Wasserman, N., 2016. The throne vs. the kingdom: founder control and value creation in startups. *Strateg. Manag. J.* Forthcoming).
- Whitley, R., 1999. *Divergent Capitalisms: The Social Structuring and Change of Business Systems*. Oxford University Press, New York.
- Wooldridge, J., 2012. *Introductory Econometrics: A Modern Approach*. fifth ed. Cengage Learning, Mason, OH.
- Wright, M., Filatotchev, I., Hoskisson, R.E., Peng, M.W., 2005. Strategy research in emerging economies: challenging the conventional wisdom. *J. Manag. Stud.* 42, 1–33.
- Wright, M., Hmieleski, K.M., Siegel, D.S., Ensley, M.D., 2007. The role of human capital in technological entrepreneurship. *Entrep. Theory Pract.* 31, 791–806.
- Xia, J., Boal, K., Delios, A., 2009. When experience meets national institutional environmental change: foreign entry attempts of U.S. firms in the Central and Eastern European region. *Strateg. Manag. J.* 30, 1286–1309.
- Xin, K.R., Pearce, J.L., 1996. Guanxi: connections as substitutes for formal institutional support. *Acad. Manag. J.* 39, 1641–1658.
- Xu, D., Meyer, K.E., 2013. Linking theory and context: "strategy research in emerging economies" after Wright et al. (2005). *J. Manag. Stud.* 50, 1322–1346.
- Yang, J.Y., Li, J., 2008. The development of entrepreneurship in China. *Asia Pac. J. Manag.* 25, 335–359.
- Yu, J., Stough, R.R., 2005. The determinants of entrepreneurship development in China. *Int. J. Manag. Ent. Dev.* 3, 30–52.
- Zhou, W., 2009. Bank financing in China's private sector: the payoffs of political capital. *World Dev.* 37, 787–799 (Law, Finance and Economic Growth in China).
- Zhou, W., 2013. Political connections and entrepreneurial investment: evidence from China's transition economy. *J. Bus. Ventur.* 28, 299–315.
- Zhou, W., 2014. Regional institutional development, political connections, and entrepreneurial performance in China's transition economy. *Small Bus. Econ.* 43, 161–181.
- Zhou, W., 2016. Institutional environment, public-private hybrid forms, and entrepreneurial reinvestment in a transition economy. *J. Bus. Ventur.* <http://dx.doi.org/10.1016/j.jbusvent.2016.11.002>.